Supplemental Materials

System identification of robotic fish swimming reveals roles of reactive and resistive fluid force in two distinct swimming gaits

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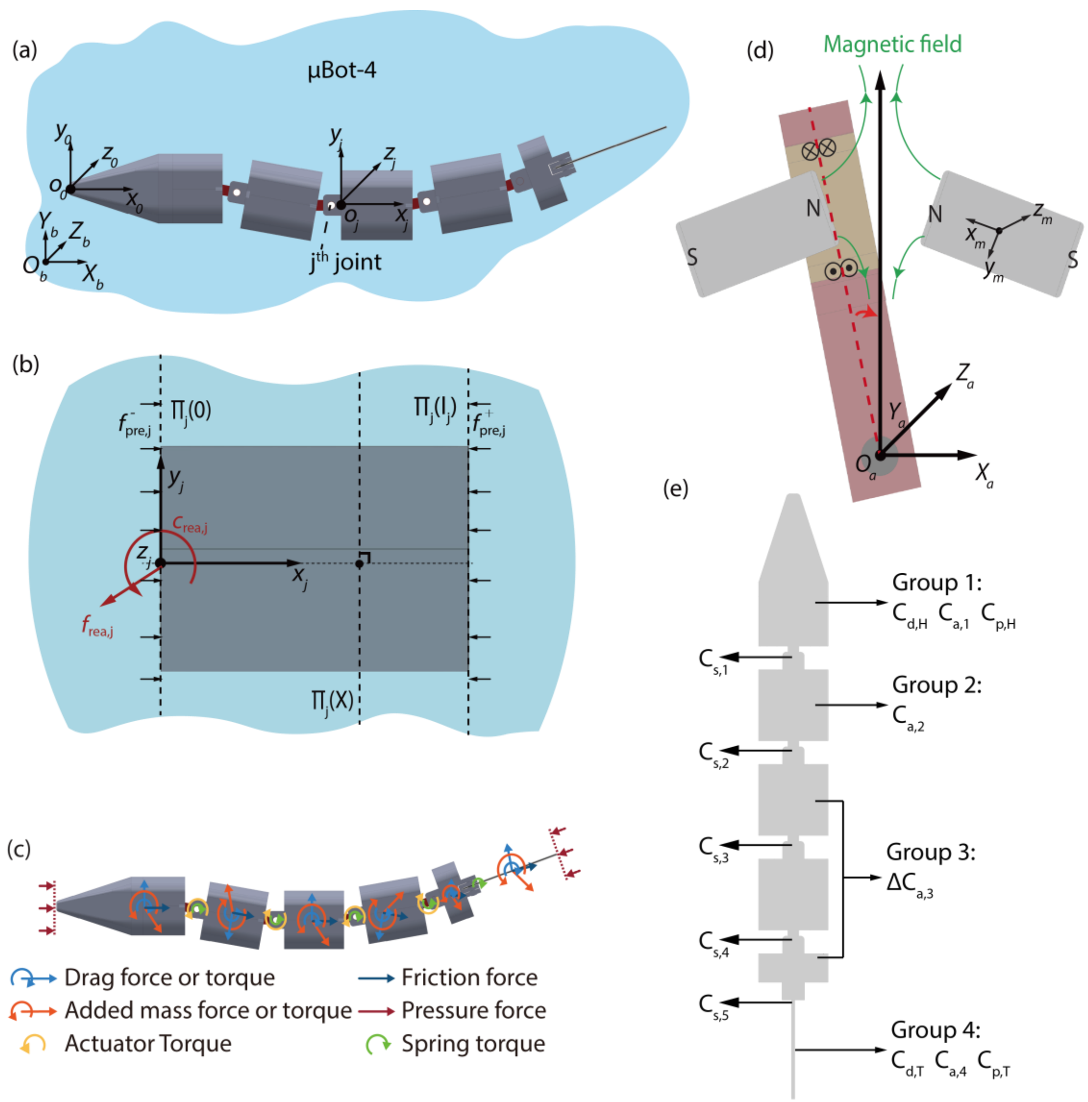

Figure S1. Details of the MBot-4 modeling. (a) The coordinate frame attached on mBot-4. (b) The added-mass volume (AMV). (c) The forces and torques applied on each segments. (d) The model of electromagnetic actuator. (e) The sub-groups and fitted modeling parameters.

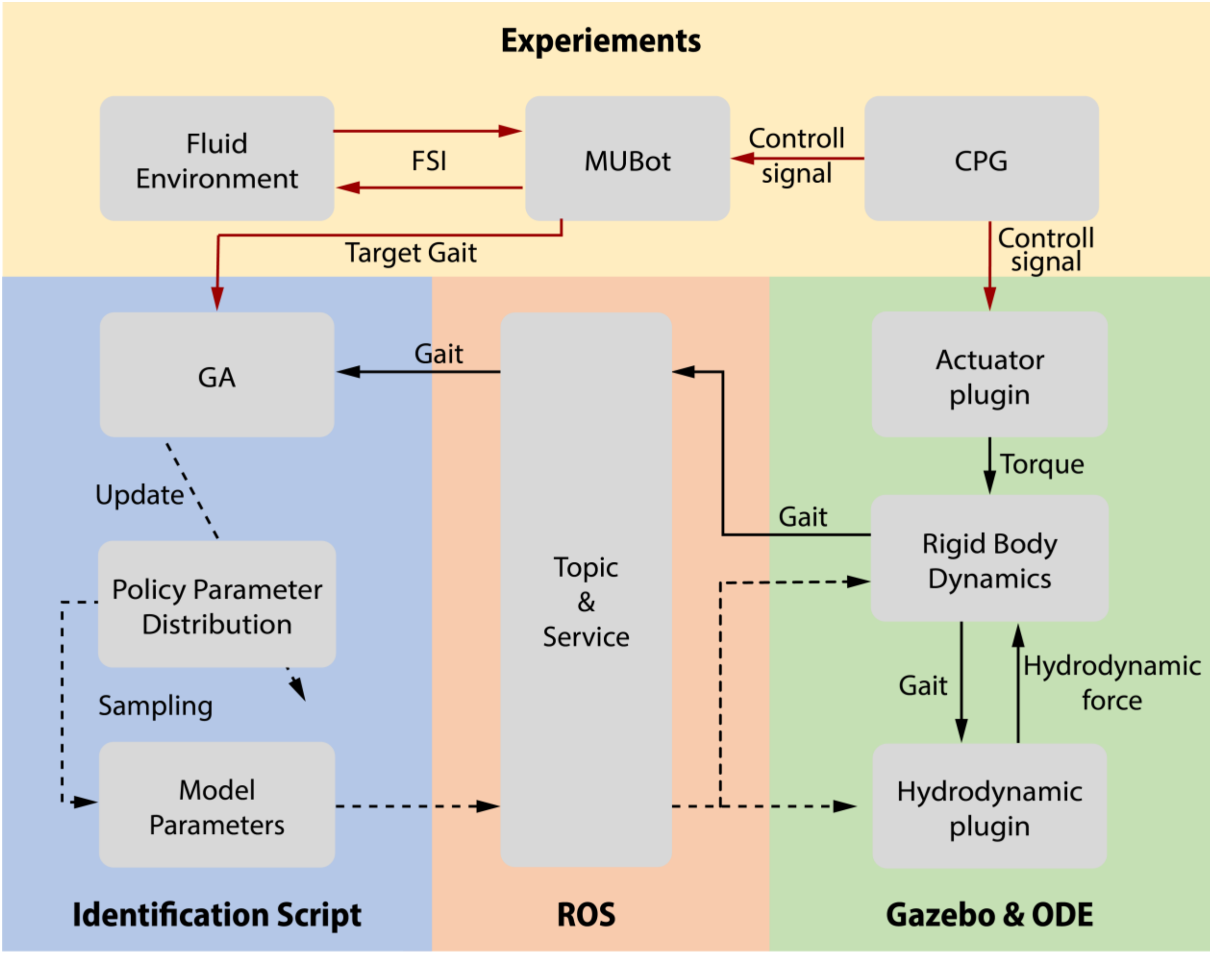

Figure S2. Flow chart for implementation of the system identification with ROS and Gazebo simulation.

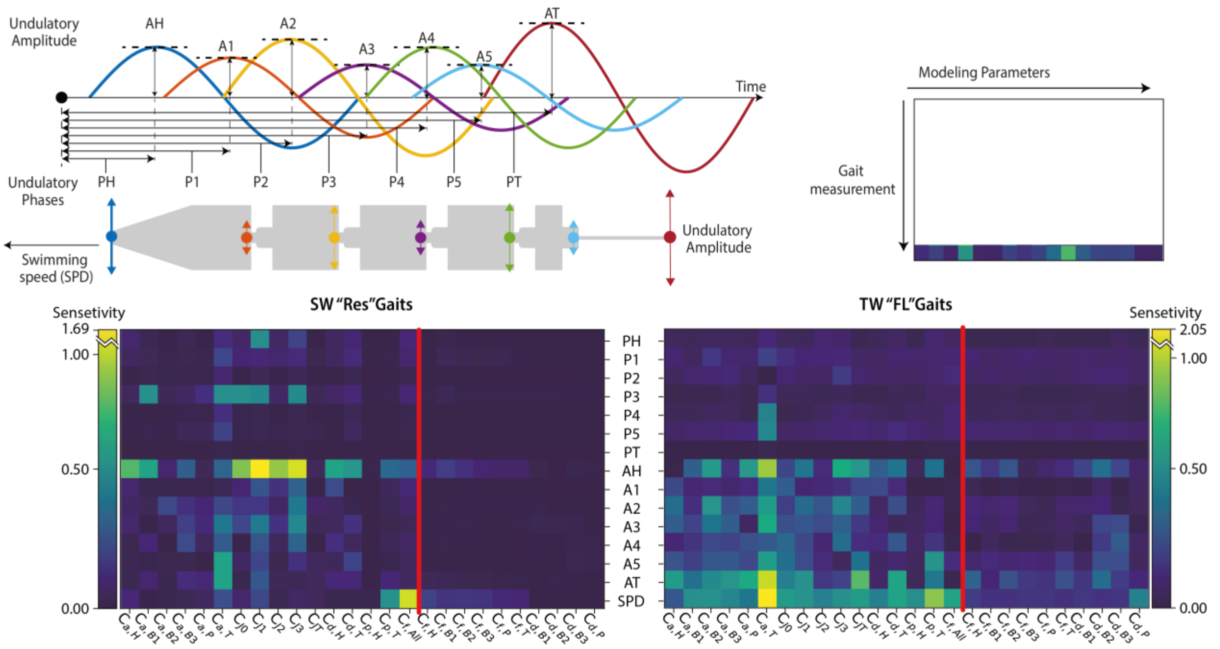

Figure S3. Perturbation test of μBot-4 under SW "Res" Gaits and TW "FL" Gaits. Gait parameters named as AH to AT denote the undulation amplitude in lateral direction and PH to PT denote the phase of undulation of marker. Modeling parameters named with H, B1, B2, B3, P and T denote head segment, first body segment to third body segment, peduncle and caudal fin. $C_{f,all}$ denotes the friction coefficients applied on all segments and perturbed at the same time. On the left side of the red line, these are the modeling parameters that are fitted in the process.

Table S1 Mechanical properties of MUBot-4

	Mass(g)	Length(mm)	Spring stiffness (Nmm/rad)
Head segment	9.96	43.00	5.33
Body segment(s)	10.68	26.75	5.33
Peduncle	7.51	19.75	5.33
Caudal fin	0.53	28.00	33.71

Table S2 Subset of segments and parameters for μBot-4

	μBot-4
Group 1	Head segment
Fitted parameter	$C_{d,H}, C_{a,1}, C_{p,H}$
Group 2	Body segment 1
Fitted parameter	$C_{a,2}$
Fixed parameter	$C_d = 2.25$
Group 3	Body segments 2, Peduncle segment
Fitted parameter	$\Delta C_{a,3}$
Fixed parameter	$C_d = 2.25$
Group 4	Tail segment
Fitted parameter	$C_{d,T}, C_{a,4}, C_{p,T}$
Universal fitted parameter	C_f

Table S3. Bounds and deviation of parameters for system identification and perturbation test.

Parameters	Bounds	Deviation
C_d	$[\ \ 1.5, 4.5]$	0.50
C_a	$[\ \ 0.0, 1.0]$	0.20
ΔC_a	$[-0.1, 0.1]$	0.05
C_p	$[\ \ 0.0, 1.0]$	0.20
C_s	$[-0.3,0.3]$	0.05

Table S4 Objective function and relative errors for two gaits

	SW “Res” Gaits	TW “FL” Gaits
E_g	0.200	0.370
E_{vel}	0.00681	0.061
E_{gait}	0.296	0.464

Supplementary Equations

To describe the motion of μBot-4, we placed 7 markers (shown in Figure S2) on the μBot-4 in both experiments and simulations, and fitted the lateral displacement of the markers as sinusoidal wave as following

$$G_i(t) = A_i \sin\big(2\pi(f \cdot t + \Phi_i)\big) \quad \#SE(1)$$

$$A_i, \Phi_i = \arg\min \Sigma_t\big(Y_i(t) - G_i(t)\big), \#SE(2)$$

where $G_i(t)$denotes gaits functions for i-th marker , A_i, Φ_i are the fitted amplitude and phase for the i-th body points, f is the frequency of input signal and $Y_i(t)$ is the lateral position of i-th marker.

Supplemental Movie Captions:

Movie S1: Animated comparison between experimentally-measured gaits and the simulation-predicted gaits, including both traveling-wave "fish-like" (TW "FL") gait and a standing-wave "resonant" (SW "Res") gait.

Movie S2: Wake structures of μbot measured by Particle Image Velocimetry (PIV) for both traveling-wave "fish-like" (TW "FL") gait and a standing-wave "resonant" (SW "Res") gait.